

# Synthetic Image Generation

[https://github.com/aiyshwary/Synthetic\\_waste\\_image\\_generation](https://github.com/aiyshwary/Synthetic_waste_image_generation)

## OVERVIEW

An automated pipeline for generating high-quality synthetic training images for object detection, addressing compositing annotated object crops onto varied backgrounds with randomized augmentations.

## IMPACT

Built an end-to-end synthetic image generation pipeline that composites real object crops onto diverse backgrounds with randomized augmentations, producing annotated training data that reduces dependency on costly manual data collection.

## KEY DETAILS

1. Extracts foreground object crops from existing annotated datasets using bounding box and mask information.
2. Composites crops onto varied background images with randomized scale, rotation, and position transformations.
3. Automatically generates YOLO/Pascal VOC-compatible annotation files alongside each synthetic image.
4. Targets underrepresented classes to balance dataset distributions for improved model generalization.
5. Validated improvements in downstream detection model performance on augmented versus original-only data.
6. Uses Ultralytics `auto_annotate` with a custom YOLO detection model and SAM2 for automatic polygon annotation.
7. Implements three synthesis strategies: CropScatterSynth (up to 30 objects/image), ClusteredSynthetic, and SyntheticPhaseGen (three-phase YAML-configurable generation) with multithreaded generation.
8. Dockerized on `nvidia/cuda:12.2.0` with full 9-step bash pipeline: S3 download, annotation conversion, background selection, synthetic generation, and visualization.